Design Engineering Manager
Location: Sharjah
Salary: AED 18,000 – 26,000

Industry: Water Treatment

Job Brief:

We are seeking an experienced Design Engineering Manager in Sharjah to lead the full engineering design lifecycle for major water and wastewater infrastructure projects, including desalination plants, sewage treatment plants, ZLD systems, and pumping stations.

The role requires a technically strong leader capable of managing multidisciplinary engineering teams, ensuring high-quality design deliverables, and aligning engineering outputs with procurement and execution requirements.

Job Responsibilities:

A. Design Leadership

- Lead and manage the complete design lifecycle from Concept → Basic → Detailed Engineering.
- Review, validate, and approve all engineering deliverables, including P&IDs, hydraulic calculations, layouts, SLDs, BOQs, technical specifications, and data sheets.
- Ensure design alignment with client requirements, project specifications, and authority standards.
- Provide technical solutions for complex design challenges and cross-disciplinary issues.

- Maintain high standards of detail, accuracy, and constructability in all deliverables.

B. Detailed Engineering Planning & Forecasting (Critical Responsibility)

This is one of the core KPIs for the role.

1. Work Breakdown Structure (WBS)

- Prepare a complete and detailed Work Breakdown Structure for every project.
- Break down work into discipline-wise and package-wise tasks to ensure clarity and accountability.

2. Engineering Deliverables Planning

- Define all deliverables required for Concept, Basic, and Detailed Engineering phases.
- Develop and maintain a comprehensive engineering deliverable register with defined timelines.
- Track progress on a daily and weekly basis to ensure zero delays in deliverables.

3. Forecasting & Resource Allocation

- Estimate the engineering hours required for each discipline.
- Assign tasks across Mechanical, Electrical, Process, Civil (vendor), and drafting teams.
- Identify bottlenecks early and implement corrective actions to maintain progress.

4. Coordination with Execution Team

- Provide critical construction inputs, including foundation loads, cable routing, piping layouts, method statements, and installation drawings.
- Support procurement by ensuring the timely delivery of data sheets, specifications, and vendor approvals.
- Align engineering milestones with procurement schedules and site execution timelines.

5. Monitoring & Control

- Conduct weekly engineering review meetings to track progress and address issues.
- Maintain dashboards reflecting percentage progress, pending items, risks, and recovery plans.
- Take ownership of resolving delays, design gaps, and coordination challenges.

6. Accountability

- Ensure zero rework due to engineering errors through stringent quality control.
- Ensure engineering deliverables do not delay procurement or site execution.
- Take full responsibility for maintaining the engineering schedule from start to completion.

C. Project Execution & Technical Support

- Support tender engineering and preparation of technical proposals.
- Participate in technical discussions with clients, consultants, and OEMs.
- Lead design reviews, HAZOP studies, value engineering, and coordination meetings.
- Provide technical support to site teams during installation, testing, commissioning, and troubleshooting.
- Review and validate vendor designs and Factory Acceptance Test (FAT) reports.

D. Team Management

- Lead and mentor a team of 10+ engineers across Mechanical, Electrical, Process, and Drafting disciplines.
- Monitor individual performance, ensuring quality output and adherence to timelines.
- Develop team capabilities through structured training and technical development initiatives.
- Ensure effective coordination and communication across all engineering disciplines.

E. Quality & Documentation

- Establish and implement internal engineering standards, templates, and checklists.
- Ensure all drawings and documents meet IFC (Issued for Construction) quality standards.
- Manage document control and versioning through engineering document management systems.
- Ensure compliance with international standards such as AWWA, ASTM, ASME, BS, IEC, and NFPA.

F. Stakeholder Coordination

- Coordinate with clients, consultants, technology licensors, and government authorities.
- Present design concepts and technical solutions during workshops and approval meetings.
- Support procurement teams in vendor evaluation and finalization processes.

G. Required Technical Expertise

- Strong hands-on design experience in:
- SWRO Plants: intake systems, pre-treatment, membrane design, ERDs, post-treatment.
- STPs: MBBR, SBR, MBR, IFAS, tertiary treatment, and sludge systems.
- ZLD Systems: high-TDS treatment, evaporators, crystallizers.
- Foul Sewage & Stormwater Pumping Stations: pump sizing, surge analysis, hydraulics.
- Hydraulics: pipe sizing, head loss calculations, surge analysis, pump selection.
- Electrical systems: load analysis, SLDs, MCC design, cable sizing.
- Instrumentation: instrument selection, IO lists, control philosophy, SCADA fundamentals.
- International codes and standards: AWWA, ASTM, ASME, BS, IEC, NFPA, DEWA/ADDC/SEWA guidelines.

Job Requirements:

- Bachelor’s degree in **Mechanical, Electrical, or Chemical Engineering.**
- Master’s degree is preferred.
- Minimum **15+ years** of experience in **water and wastewater design.**
- At least **5+ years** of experience in a **Design Manager or Senior Engineering Manager role.**
- Highly detail-oriented with a methodical approach to engineering design.
- Strong planning and forecasting capabilities, especially in engineering deliverables.
- Hands-on design expertise with deep technical knowledge.
- Excellent communication, coordination, and documentation skills.
- Strong leadership, mentoring, and decision-making abilities.
- Ability to work under pressure and meet strict project deadlines.

KPIs for Performance Evaluation:

- Timely submission of all engineering deliverables.
- High quality and accuracy of work with minimal or zero rework.
- Reliability and accuracy of WBS and engineering time forecasts.
- Efficiency and performance of the design team.
- Turnaround time for client and consultant approvals.
- Effective coordination with procurement and project execution teams.